Metro Ticketing System using ServiceNow - Capstone Project Documentation

1. Introduction

- **Project Title:** Metro Ticketing System using ServiceNow.

2. Project Overview

Purpose

The Metro Ticket Generating System is developed to digitize and automate the metro ticket booking and validation process using the ServiceNow platform. The system aims to eliminate manual ticketing procedures, reduce passenger waiting time, and provide a secure, efficient, and contactless travel experience.

Features

The key features of the Metro Ticket Generating System include:

- Online metro ticket booking through Service Portal
- Selection of source and destination metro stations
- Automatic fare calculation based on journey details
- Support for single and return journey types
- Passenger count management
- Unique ticket number generation
- QR code generation for digital tickets
- Ticket validation using QR codes
- Centralized metro ticket database management
- Automated notifications and request processing
- Reports and dashboards for monitoring and analytics

The system leverages ServiceNow components such as Catalog Items, Catalog Client Scripts, Business Rules, Script Includes, Flow Designer, UI Pages, and custom database tables to provide a complete digital metro ticketing solution.

3. Architecture

The Metro Ticket Generating System follows a layered architecture approach on the ServiceNow platform to provide a secure, scalable, and automated ticketing solution. The architecture is divided into Presentation Layer, Application Logic Layer, Data Layer, and Integration & Analytics Layer.

1. Presentation Layer (User Interface)

The Presentation Layer is implemented using ServiceNow Service Portal, Catalog Items, and UI Pages. It provides a user-friendly interface where passengers can book metro tickets, view generated tickets, and validate QR-based tickets. Catalog Client Scripts and UI Policies are used to perform real-time validations and improve user experience.

2. Application Logic Layer (Backend Processing)

The Application Logic Layer handles the core business operations of the system using Business Rules, Script Includes, and Flow Designer. This layer is responsible for fare calculation, ticket number generation, QR code generation, request processing, workflow automation, and ticket validation.

3. Data Layer (Database Management)

The Data Layer uses ServiceNow custom tables to store and manage metro station information, ticket booking details, passenger information, fare data, and generated ticket records. The centralized database ensures efficient storage, retrieval, and management of metro ticket information.

4. Integration & Analytics Layer

The Integration Layer uses external QR code generation APIs to create digital QR tickets. Notifications, reports, and dashboards provide automated communication, ticket tracking, booking statistics, and analytical insights for administrators and users.

The layered architecture ensures modularity, maintainability, scalability, and secure management of the Metro Ticket Generating System while providing an efficient digital ticketing experience.

4. Setup Instructions

Prerequisites

The following software components and services are required to set up and run the Metro Ticket Generating System:

- ServiceNow Developer Instance
- ServiceNow Studio
- Service Portal
- Flow Designer
- Catalog Builder

- Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)
- Internet Connectivity
- QR Code Generation API Service
- ServiceNow Administrator Access

Installation and Configuration

Step 1: Create ServiceNow Developer Instance

- Register and activate a ServiceNow Developer Instance.
- Log in with administrator credentials.

Step 2: Create Application

- Open ServiceNow Studio.
- Create a new application named **Metro Ticket Generating System**.

Step 3: Create Database Tables

- Create the **Metro Station** table.
- Create the **Metro Database** table.
- Configure all required fields and relationships.

Step 4: Create Catalog Items

- Create the **Book QR Ticket** catalog item.
- Create the **Scan Metro Ticket** catalog item.
- Add all required variables and form fields.

Step 5: Configure Business Logic

- Create Catalog Client Scripts for validations and fare calculations.
- Create Business Rules for ticket generation and processing.
- Create Script Includes for reusable server-side logic.
- Configure Flow Designer workflows for automation.

Step 6: Configure User Interface

- Configure Service Portal pages.

- Create UI Pages for ticket display and validation.
- Configure reports and dashboards.

Step 7: Configure Integrations

- Integrate the QR Code Generation API.
- Configure email notifications and automated alerts.

Step 8: Testing and Validation

- Test ticket booking functionality.
- Verify fare calculation.
- Validate QR code generation.
- Test ticket validation workflow.
- Verify database records and notifications.

5. Folder Structure

Since the Metro Ticket Generating System is developed on the ServiceNow platform, the project follows a component-based architecture.

Application Components Structure

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Metro Ticket Generating System

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├── Service Portal

│   ├── Book QR Ticket

│   └── Scan Metro Ticket

│

├── Catalog Items

│   ├── Book QR Ticket

│   └── Scan Metro Ticket

│

├── Catalog Client Scripts

│
```

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├── UI Policies
│
├── Business Rules
│
├── Script Includes
│
├── Flow Designer
│   └── Ticket Processing Flow
│
├── UI Pages
│   └── Ticket Display Page
│
├── Database Tables
│   ├── Metro Station Table
│   └── Metro Database Table
│
├── Notifications
│
├── Reports
│
└── Dashboards
```

Service Portal Layer

Contains the user-facing interfaces for metro ticket booking, QR ticket generation, and ticket validation.

Application Logic Layer

Includes Catalog Client Scripts, Business Rules, Script Includes, UI Policies, and Flow Designer workflows responsible for validation, fare calculation, ticket generation, and automation.

Data Management Layer

Consists of custom ServiceNow tables that store metro station information, ticket records, passenger details, and QR ticket data.

Analytics and Notification Layer

Includes notifications, reports, and dashboards used for ticket confirmation, monitoring, analytics, and system reporting.

6. Running the Application

The Metro Ticket Generating System runs on the ServiceNow platform through the Service Portal and automated backend workflows.

Step 1: Access the Service Portal

Users log in to the ServiceNow instance and navigate to the Metro Ticket Generating System Service Portal.

Step 2: Book Metro Ticket

The user opens the **Book QR Ticket** catalog item and enters the required journey details:

- Source Station
- Destination Station
- Number of Passengers
- Journey Type (Single/Return)
- Payment Mode

Step 3: Validate User Input

Catalog Client Scripts and UI Policies validate the entered information and automatically calculate the fare based on the selected route.

Step 4: Submit Ticket Request

After successful validation, the user submits the ticket booking request. A request record and unique ticket number are generated automatically.

Step 5: Backend Processing

Business Rules, Script Includes, and Flow Designer workflows process the request by:

- Calculating the final fare
- Generating the ticket number

- Creating the QR code
- Storing ticket information in the Metro Database

Step 6: Generate Digital Ticket

The system generates a digital metro ticket containing:

- Ticket Number
- Journey Details
- Passenger Information
- Fare Amount
- QR Code

Step 7: Ticket Validation

The user accesses the **Scan Metro Ticket** module to validate the generated ticket using the ticket number or QR code.

Step 8: Monitoring and Reporting

Administrators can monitor ticket bookings, validation status, and operational statistics through ServiceNow reports and dashboards.

7. API Documentation

The Metro Ticket Generating System primarily utilizes ServiceNow's internal automation framework, including Business Rules, Script Includes, Flow Designer, and Notifications. Additionally, an external QR Code Generation API is integrated to create digital metro tickets.

Internal ServiceNow Automation

The following operations are handled internally within the ServiceNow platform:

- Metro ticket booking request processing
- Fare calculation based on selected stations
- Ticket number generation
- Request and ticket record creation
- Workflow automation using Flow Designer
- Ticket validation processing
- Notification generation and delivery

External QR Code Generation API

The system integrates an external QR Code Generation API to generate QR-based digital metro tickets.

API Endpoint

https://api.qrserver.com/v1/create-qr-code/?size=200x200&data=<ticket_data>

Example Request

https://api.qrserver.com/v1/create-qr-code/?size=200x200&data=MT123456

API Workflow

Metro Ticket Booking

↓

Business Rule Trigger

↓

Script Include Processing

↓

QR Code API Request

↓

QR Code Generation

↓

Store QR Code in Metro Database

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Display Ticket and Send Notification

The API integration enables secure, contactless, and efficient digital ticket generation and validation within the Metro Ticket Generating System.

8. Authentication

The Metro Ticket Generating System utilizes ServiceNow's built-in authentication and Role-Based Access Control (RBAC) framework to ensure secure access to application data and functionalities.

Authentication Mechanism

User authentication is managed by the ServiceNow platform. Users must log in using valid ServiceNow credentials before accessing the Service Portal, metro ticket booking modules, and administrative functionalities. Session management and user verification are handled by ServiceNow's security framework.

Role-Based Access Control (RBAC)

The system implements role-based access to restrict functionalities based on user responsibilities.

User Role (Metro User)

- Access the Service Portal
- Book metro tickets
- View generated tickets
- Validate their ticket information

Administrator Role (Admin)

- Manage metro stations and fare details
- Configure catalog items and workflows
- Access reports and dashboards
- Monitor ticket booking and validation activities

Validator Role (Metro Validator)

- Access the ticket validation module
- Verify ticket information and QR codes
- Monitor ticket validation status

Security Controls

The following security mechanisms are implemented:

- Table-level security using Access Control Lists (ACLs)
- Role-based access restrictions
- Record-level access control

- Server-side validation through Business Rules and Script Includes
- Session-based authentication provided by ServiceNow

These authentication and authorization mechanisms ensure secure access, data integrity, and controlled operations within the Metro Ticket Generating System.

9. User Interface

The Metro Ticket Generating System provides a user-friendly and responsive interface developed using ServiceNow Service Portal, Catalog Items, and UI Pages. The interface is designed to simplify ticket booking, ticket generation, and ticket validation while ensuring a seamless user experience.

1. Metro Ticket Booking Interface

The primary interface of the system is the Book QR Ticket catalog item available in the Service Portal. Users can enter journey details including:

- Source Station
- Destination Station
- Number of Passengers
- Journey Type (Single/Return)
- Payment Mode

Catalog Client Scripts and UI Policies provide real-time validation, automatic fare calculation, and user-friendly error messages to improve the booking experience.

2. Ticket Generation Interface

After successful ticket booking, the system generates a digital metro ticket containing:

- Request Number
- Ticket Number
- Passenger Details
- Journey Information
- Fare Amount
- Generated QR Code

The ticket information is displayed through a dedicated UI Page and stored in the Metro Database for future reference.

3. Ticket Validation Interface

The Scan Metro Ticket module provides an interface for ticket validation. Users or validators can verify the ticket by entering the ticket number or scanning the QR code. The system retrieves the ticket details and displays the validation status.

4. Administrative Interface

Administrators can access custom tables, reports, and dashboards to manage:

- Metro Station Information
- Fare Details
- Ticket Records
- Booking Statistics
- Validation Reports
- System Performance Metrics

5. User Experience Design

The user interface is designed with the following principles:

- Simplicity and ease of use
- Responsive and user-friendly design
- Real-time validations and feedback
- Secure and reliable operations
- Consistent navigation and layout
- Efficient ticket booking and validation workflow

10. Testing

The Metro Ticket Generating System was tested using a structured manual testing approach to ensure the accuracy, reliability, and functionality of the ticket booking, fare calculation, QR code generation, and ticket validation processes.

Test Scope

Testing was performed on the following modules:

- Metro Ticket Booking
- Fare Calculation

- Ticket Number Generation
- QR Code Generation
- Metro Database Operations
- Ticket Validation
- Notification Delivery
- Reports and Dashboard

Test Cases and Results

Test Case 1: Metro Ticket Booking

- **Input:** Valid source station, destination station, passenger count, and journey type.
- **Expected Output:** Ticket booking request should be created successfully.
- **Result:** Pass

Test Case 2: Input Validation

- **Input:** Same source and destination stations selected.
- **Expected Output:** System should display a validation error and prevent submission.
- **Result:** Pass

Test Case 3: Fare Calculation

- **Input:** Valid metro route and passenger details.
- **Expected Output:** Fare should be calculated automatically and displayed correctly.
- **Result:** Pass

Test Case 4: Ticket Number Generation

- **Input:** Successful ticket booking request.
- **Expected Output:** A unique ticket number should be generated automatically.
- **Result:** Pass

Test Case 5: QR Code Generation

- **Input:** Successfully generated ticket record.
- **Expected Output:** QR code should be generated and attached to the ticket.

- **Result:** Pass

Test Case 6: Metro Database Storage

- **Input:** Completed ticket booking transaction.
- **Expected Output:** Ticket information should be stored correctly in the Metro Database.
- **Result:** Pass

Test Case 7: Ticket Validation

- **Input:** Valid ticket number or QR code.
- **Expected Output:** System should successfully validate and display ticket information.
- **Result:** Pass

Test Case 8: Notification Delivery

- **Input:** Successful ticket generation.
- **Expected Output:** Notification should be triggered and delivered successfully.
- **Result:** Pass

Testing Summary

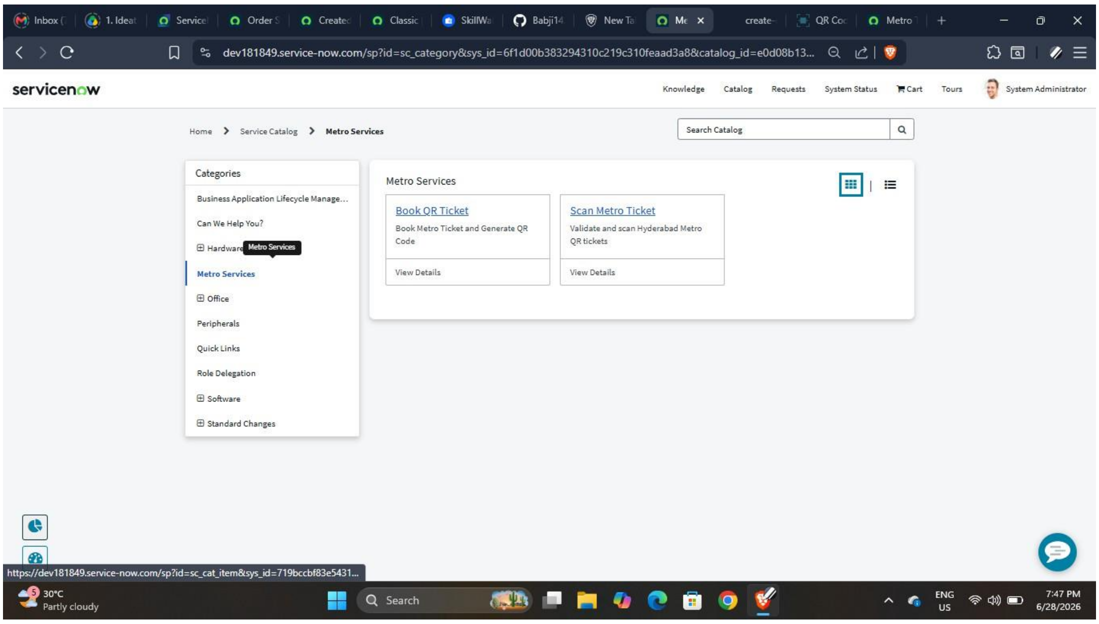
Test Module	**Status**
Ticket Booking	Pass
Input Validation	Pass
Fare Calculation	Pass
Ticket Number Generation	Pass
QR Code Generation	Pass
Metro Database	Pass
Ticket Validation	Pass
Notifications	Pass

The testing results confirm that the Metro Ticket Generating System performs reliably and satisfies the functional requirements specified during the design and development phases.

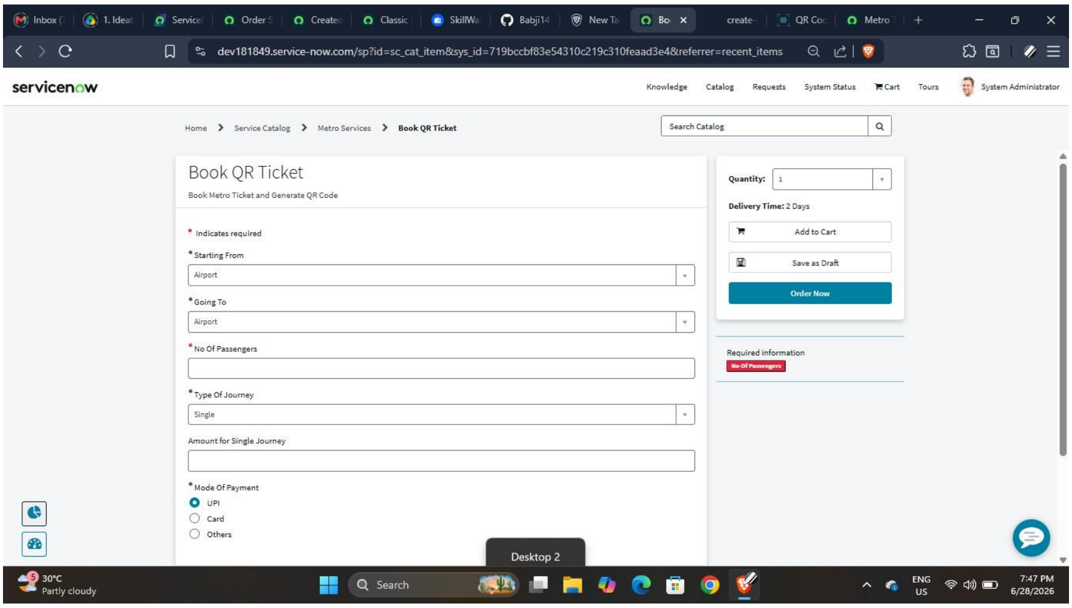
11. Screenshots or Demo

The Metro Ticket Generating System provides a user-friendly and automated digital ticketing experience through the ServiceNow platform. The following screenshots demonstrate the key functionalities and workflow of the application.

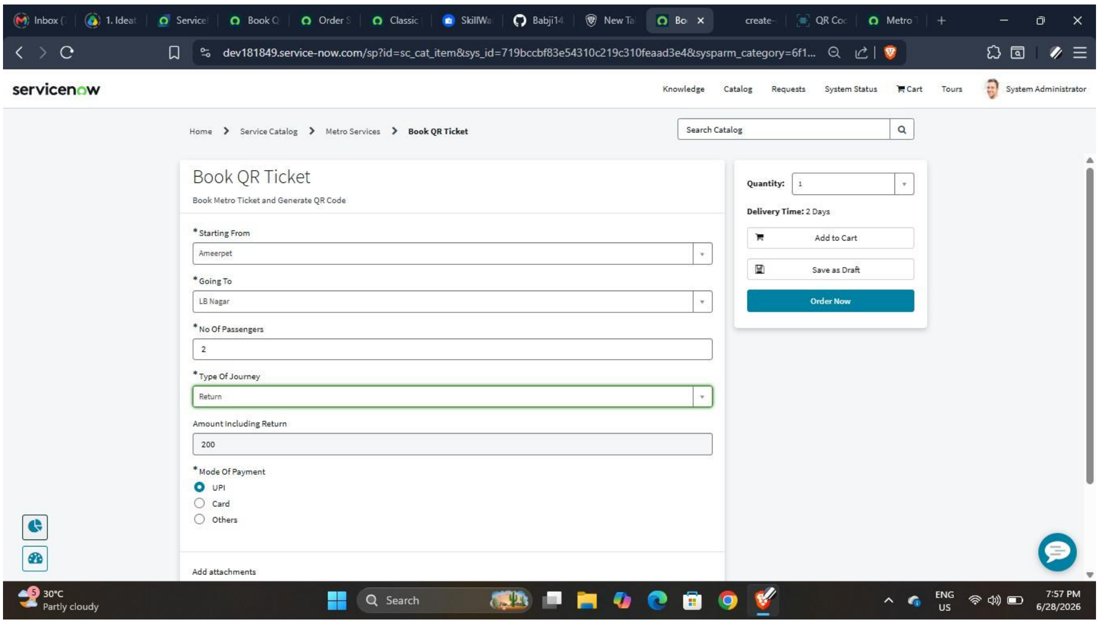
Screenshot 1: Service Portal Home Page



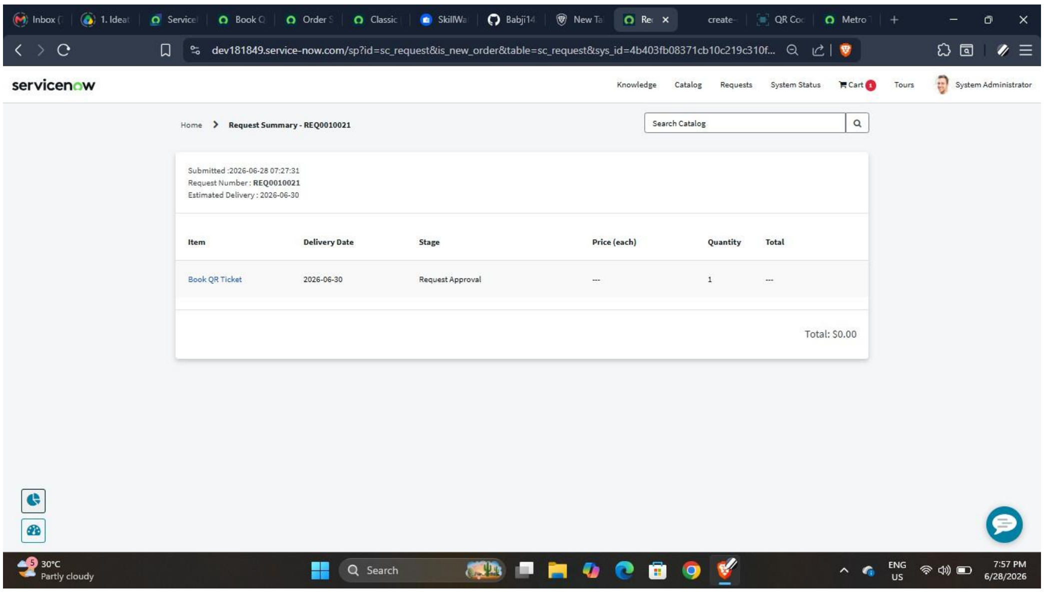
Screenshot 2: Book QR Ticket Catalog Item



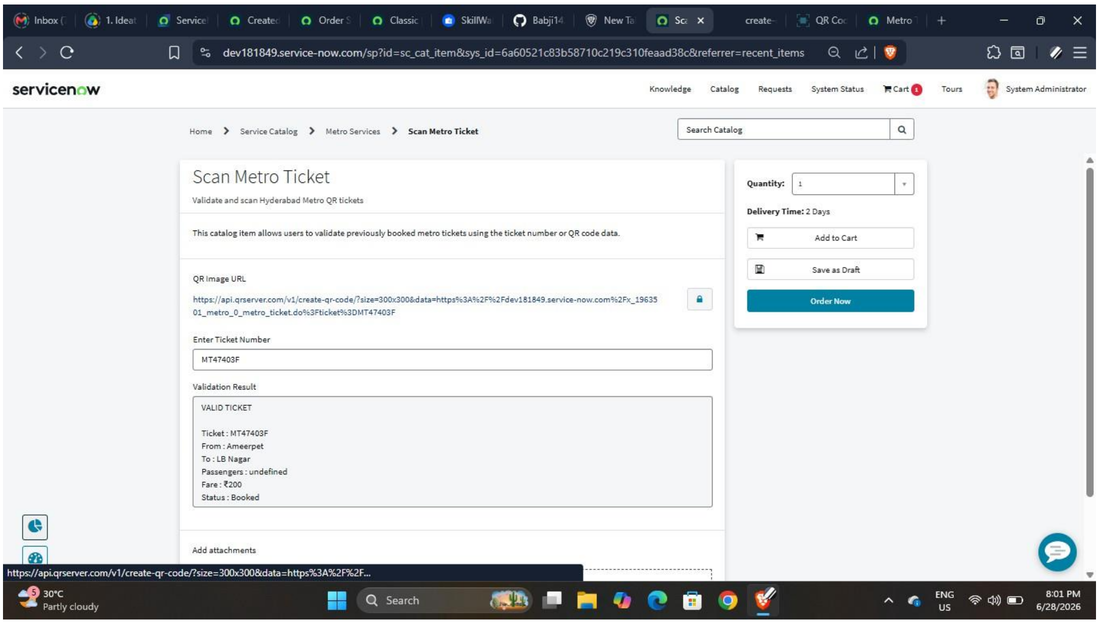
Screenshot 3: Fare Calculation and Request Submission

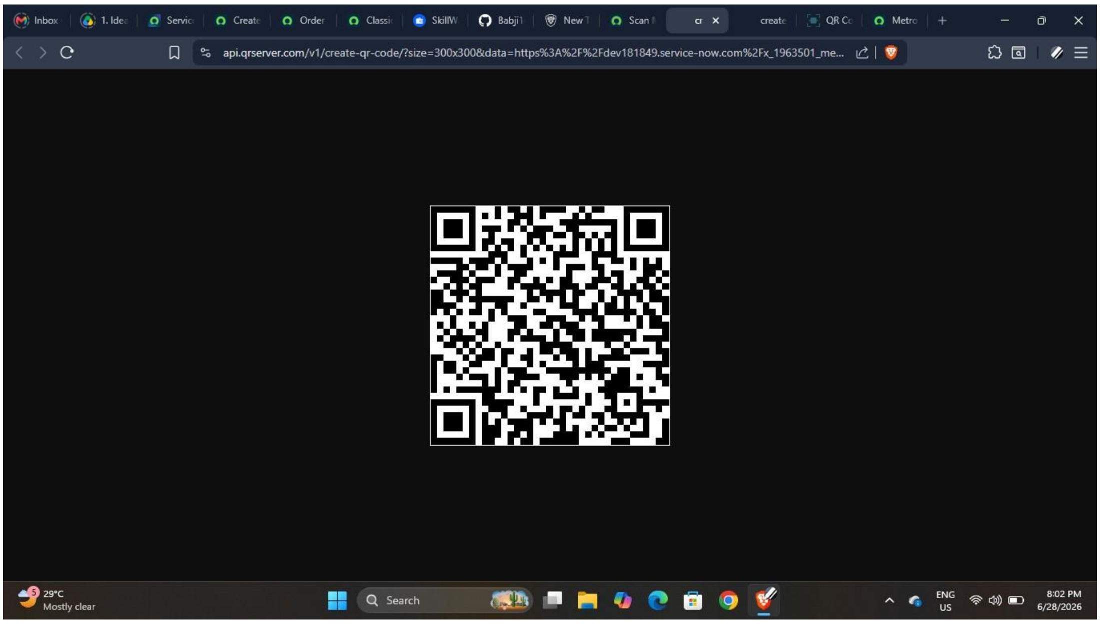


Screenshot 4: Request Summary and Ticket Number Generation

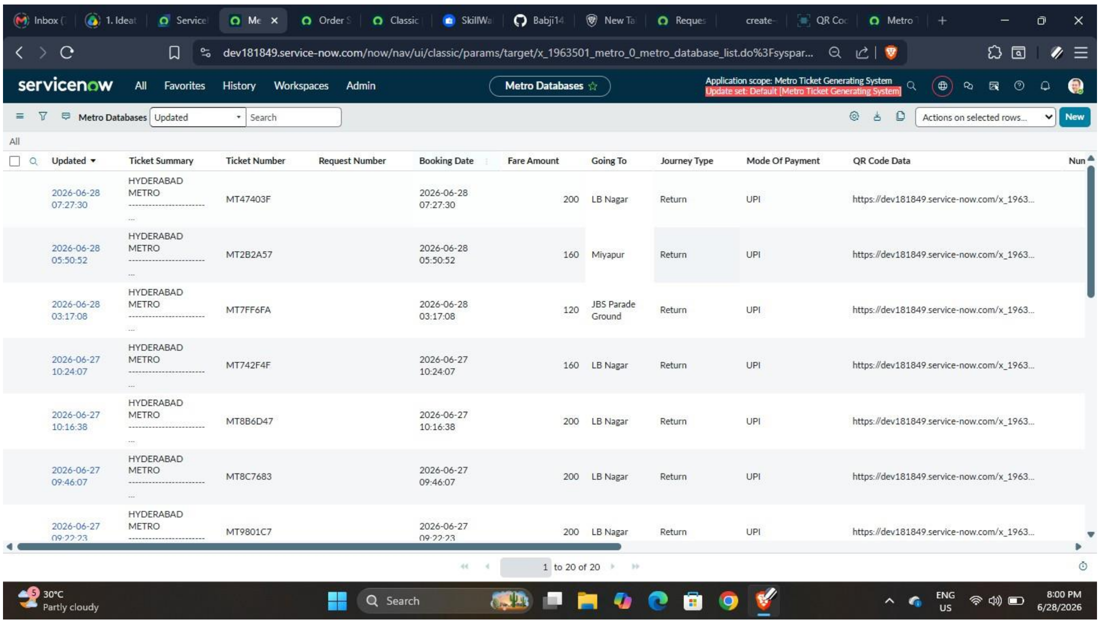


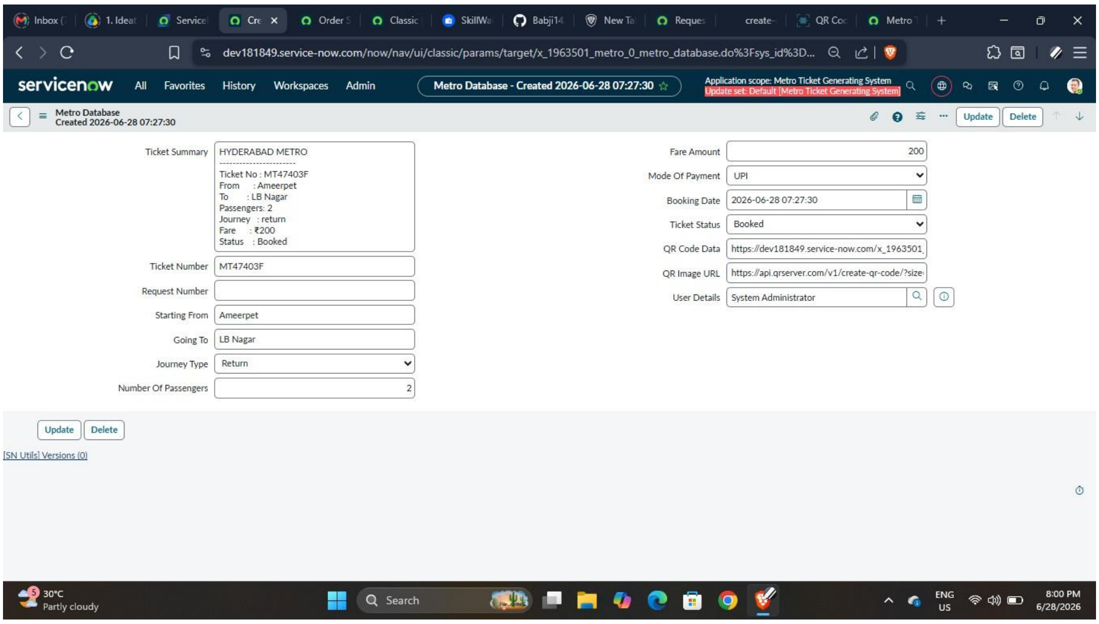
Screenshot 5: Generated Metro Ticket with QR Code



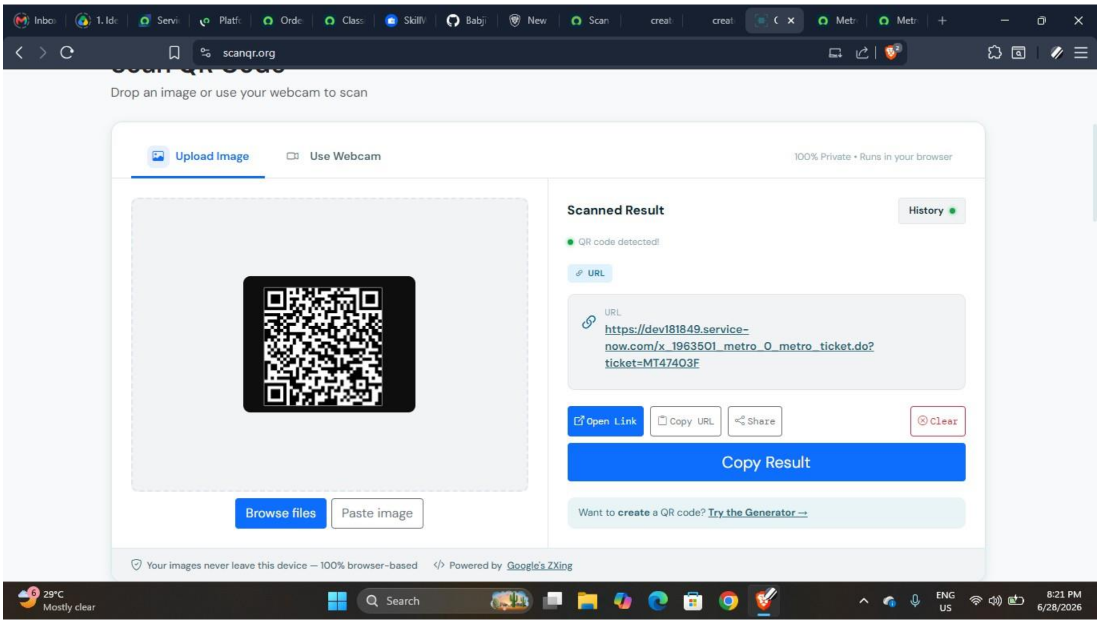


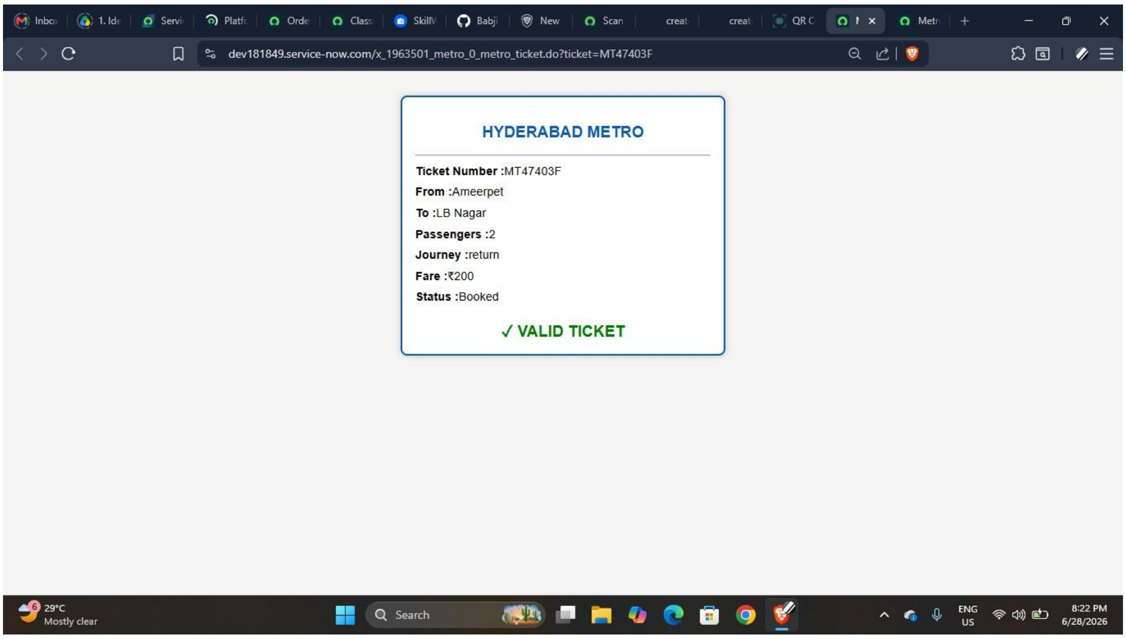
Screenshot 6: Metro Database Table





Screenshot 7: Scan Metro Ticket Interface





Demonstration Video

A project demonstration video has been created to showcase the complete workflow of the Metro Ticket Generating System, including ticket booking, fare calculation, QR code generation, ticket validation, and administrative monitoring features.

12. Known Issues

Although the Metro Ticket Generating System has been successfully implemented, certain limitations and constraints exist due to platform capabilities and external dependencies.

1. Dependency on External QR Code API

The system relies on an external QR code generation API. If the API service is unavailable or experiences delays, QR code generation and ticket confirmation may be affected.

2. No Real-Time Payment Gateway Integration

The current implementation does not support actual online payment processing. Payment mode selection is available, but financial transactions are not processed through a payment gateway.

3. Static Fare Configuration

Fare calculation is based on predefined metro station data stored in the system. Dynamic pricing based on peak hours, demand, or special events is not implemented.

4. **Limited Offline Ticket Validation**

The ticket validation process requires access to the ServiceNow platform and stored ticket records. Offline validation functionality is not currently supported.

5. Limited Scalability Testing

The application has been tested under normal usage conditions. Performance under large-scale concurrent bookings and heavy traffic scenarios has not been extensively evaluated.

6. Basic QR Security Implementation

The generated QR codes contain ticket information for validation purposes; however, advanced encryption and fraud prevention mechanisms have not been implemented in the current version.

13. Future Enhancements

The Metro Ticket Generating System can be further enhanced by incorporating advanced features and technologies to improve functionality, security, scalability, and user experience. The following enhancements are proposed for future development:

1. Online Payment Gateway Integration
Integrate secure payment gateways such as UPI, credit cards, debit cards, and digital wallets to enable real-time ticket payments.

2. Mobile Application Support
Develop Android and iOS mobile applications to provide a seamless and accessible ticket booking experience for passengers.

3. Real-Time Metro Tracking
Implement live metro train tracking and arrival prediction systems to provide real-time travel information to passengers.

4. Smart Card Integration
Integrate metro smart card systems to support automatic fare deduction and contactless travel.

5. Advanced QR Code Security
Implement encrypted QR codes and advanced validation mechanisms to improve ticket security and prevent fraud.

6. AI-Based Analytics and Reporting
Incorporate artificial intelligence and predictive analytics to analyze passenger trends, optimize operations, and generate intelligent reports.

7. Multi-City Metro Support
Extend the application to support multiple metro networks and cities within a single centralized platform.

8. Dynamic Fare Calculation
Introduce dynamic pricing models based on peak hours, demand, special events, and travel patterns.

9. Notification and Alert Enhancements

Implement SMS notifications, push notifications, and real-time alerts for booking confirmations, delays, and travel updates.

10. Cloud Scalability and Performance Optimization

Enhance the system architecture to support large-scale concurrent bookings, high availability, and improved performance for future expansion.